Statistical robustness of the reported comparisons

Supplementary note to *What Do LLM Doctors Add to Their Statistics?*

Summary

We report every comparison the paper draws a conclusion from (94 in all) with its 95% confidence interval, and repeat each without the 6 patients whose condition has no labeled patients. All numbers are produced by `tools/plan_a/robust_v1.py` from the raw interview logs.

- **The LLM's questions are useful.** Read by the statistician, they beat the same number of random questions for all five models (5.0 to 8.4 points, every interval above zero). The larger models' questions match the statistician's own: −0.2 [−4.0, +3.6].
- **Larger models with advice exceed the statistician.** At the same budget ($B = 10$): +4.9 [+1.5, +8.3]. With as many questions: +5.8 [+2.4, +9.2] at $B = 10$ and +5.6 [+2.2, +8.9] at $B = 15$. Against the statistician's own stopping rule at the same mean number of questions: +4.5 [+1.4, +7.5] and +4.7 [+1.4, +7.9]. All six per-model gains with as many questions are positive, and five are significant on their own.
- **A wrong recommendation costs the larger models little.** With the statistics shown correctly, a wrong diagnosis changes their accuracy by −2.1 [−4.0, −0.3] and wrong questions by −4.3 [−6.7, −2.0], while both stay far above no advice (+25.1 and +22.9 points).
- **Disagreement is a useful review signal.** When the larger models disagree with the statistician, they are less often right than when they agree: the gap is +36.3 [+23.5, +48.4] at $B = 10$ and +32.7 [+18.0, +47.3] at $B = 15$.
- **The conditions without labeled patients do not drive any conclusion:** excluding their 6 patients changes every difference by at most 1.5 points, and every conclusion of the paper stays the same.

1 RESULTS AT A GLANCE

Each dot is a difference in accuracy and each bar its 95% confidence interval; filled dots are significant ($p < 0.05$) and open dots are not. Dots to the right of zero favour the first condition.

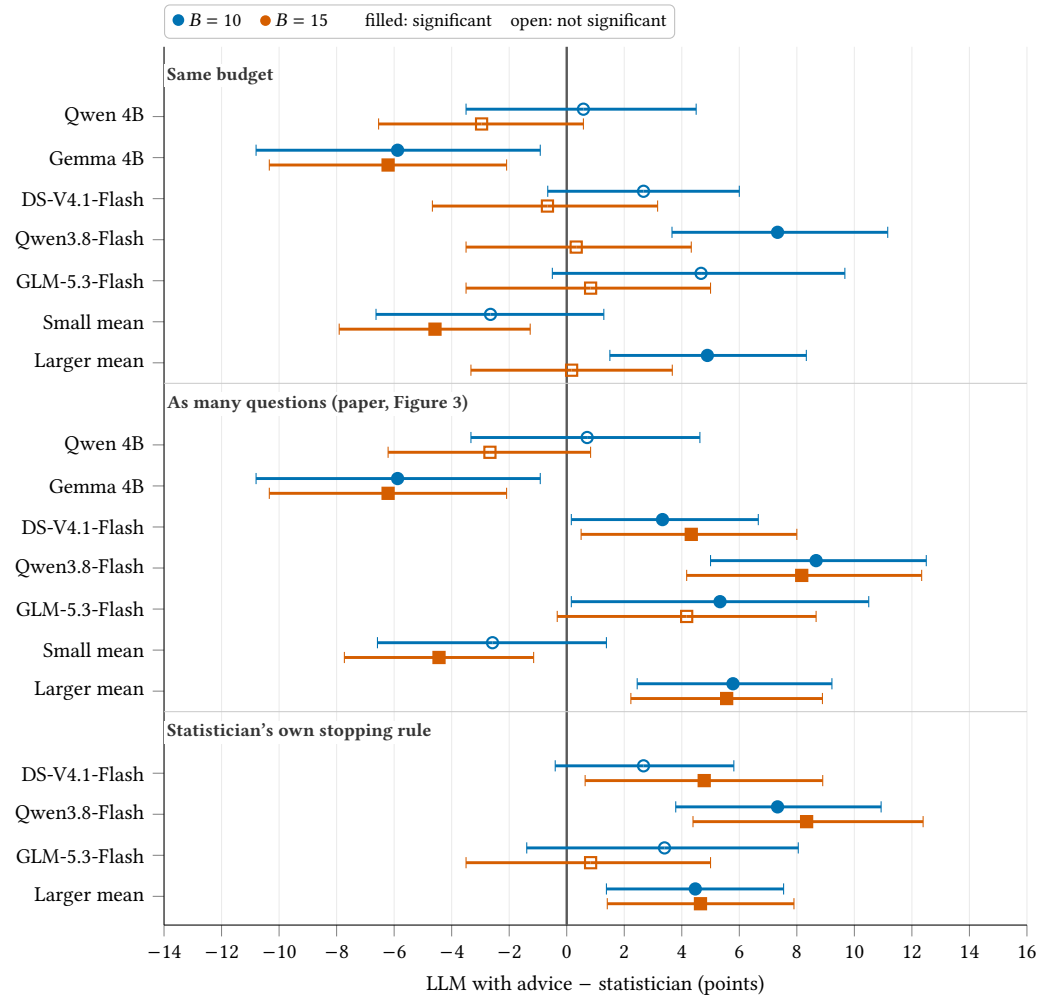


Figure 1. LLMs with advice against the statistician (paper, Table 1 and Figure 3): at the same budget, asking as many questions in each interview, and with the statistician's own stopping rule at the same mean number of questions.

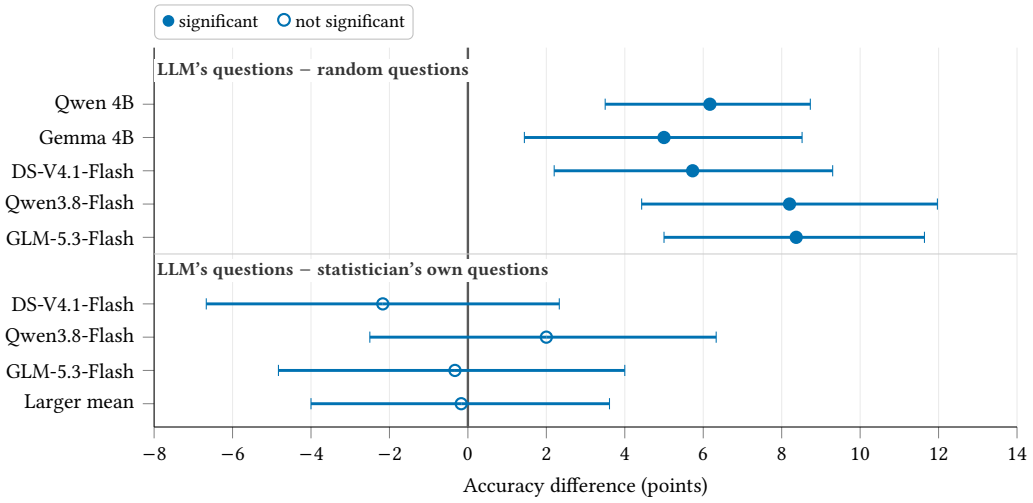


Figure 2. What the LLM's questions are worth (paper, Figure 2; $B = 15$, no advice). In every case the statistician makes the diagnosis; only the questions differ. Top: the questions the LLM chose against the same number of random questions; to the right of zero, the LLM's questions are better. Bottom: the questions the LLM chose against the same number of the statistician's own information-gain questions; near zero, they are equally good.

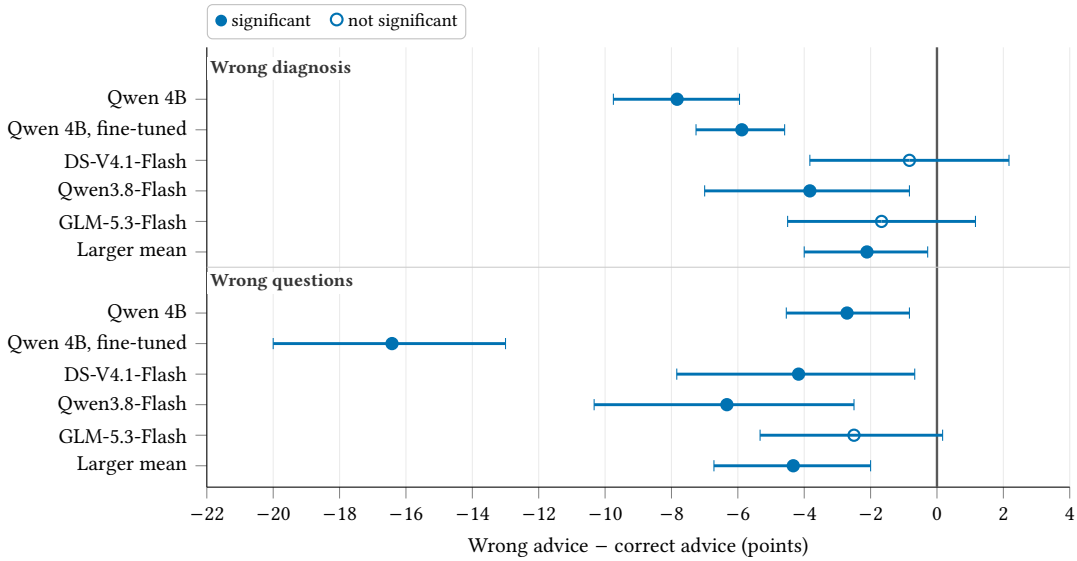


Figure 3. One wrong recommendation, with the statistics shown correctly (paper, Table 2; $B = 15$).

2 FULL RESULTS

					Without unlabeled conditions	
Comparison	Diff.	95% CI	p	MDE	Diff.	95% CI
RQ1: LLM's questions, read by the statistician ($B = 15$, no advice)						

Comparison	Diff.	95% CI	p	MDE	Diff.	95% CI
vs. random questions						
Qwen 4B	**+6.2**	[+3.5, +8.7]	<0.001	3.7	**+6.3**	[+3.6, +8.9]
Gemma 4B	**+5.0**	[+1.4, +8.5]	0.007	5.1	**+5.1**	[+1.5, +8.7]
DS-V4.1-Flash	**+5.7**	[+2.2, +9.3]	0.001	5.1	**+5.9**	[+2.2, +9.4]
Qwen3.8-Flash	**+8.2**	[+4.4, +12.0]	<0.001	5.4	**+8.4**	[+4.5, +12.1]
GLM-5.3-Flash	**+8.4**	[+5.0, +11.6]	<0.001	4.7	**+8.5**	[+5.1, +11.9]
vs. the statistician's own questions						
DS-V4.1-Flash	−2.2	[−6.7, +2.3]	0.359	6.4	−2.2	[−6.8, +2.2]
Qwen3.8-Flash	+2.0	[−2.5, +6.3]	0.398	6.3	+2.0	[−2.6, +6.5]
GLM-5.3-Flash	−0.3	[−4.8, +4.0]	0.911	6.3	−0.3	[−4.8, +4.1]
Larger mean	−0.2	[−4.0, +3.6]	0.921	5.4	−0.2	[−4.0, +3.7]
vs. the LLM's own diagnosis						
Qwen 4B	**+37.8**	[+33.1, +42.7]	<0.001	6.9	**+38.8**	[+33.9, +43.6]
Gemma 4B	**+51.8**	[+46.1, +57.4]	<0.001	8.1	**+52.8**	[+47.2, +58.5]
DS-V4.1-Flash	**+25.3**	[+19.7, +30.8]	<0.001	8.0	**+26.4**	[+20.7, +32.0]
Qwen3.8-Flash	**+23.0**	[+16.8, +29.3]	<0.001	8.9	**+24.1**	[+17.9, +30.3]
GLM-5.3-Flash	**+24.0**	[+18.5, +29.7]	<0.001	8.0	**+25.0**	[+19.4, +30.8]
RQ2: advice − no advice						
$B = 10$						
Qwen 4B	**+34.1**	[+29.4, +38.8]	<0.001	6.7	**+34.9**	[+30.1, +39.5]
Gemma 4B	**+43.5**	[+37.8, +49.3]	<0.001	8.3	**+44.4**	[+38.6, +50.3]
DS-V4.1-Flash	**+29.3**	[+23.5, +35.3]	<0.001	8.4	**+30.3**	[+24.3, +36.2]
Qwen3.8-Flash	**+28.8**	[+22.8, +34.7]	<0.001	8.6	**+29.9**	[+23.8, +36.1]
GLM-5.3-Flash	**+26.3**	[+21.2, +31.5]	<0.001	7.3	**+27.2**	[+22.1, +32.5]
$B = 15$						
Qwen 4B	**+38.5**	[+33.8, +43.2]	<0.001	6.6	**+39.4**	[+34.7, +44.1]
Gemma 4B	**+50.6**	[+45.0, +56.2]	<0.001	8.0	**+51.7**	[+46.2, +57.2]
DS-V4.1-Flash	**+28.8**	[+23.0, +34.7]	<0.001	8.3	**+29.9**	[+24.1, +35.9]
Qwen3.8-Flash	**+26.3**	[+20.5, +32.3]	<0.001	8.4	**+27.0**	[+21.1, +33.0]
GLM-5.3-Flash	**+26.5**	[+21.3, +31.7]	<0.001	7.5	**+27.4**	[+22.3, +32.8]
RQ2: LLM with advice − statistician, same budget						
$B = 10$						
Qwen 4B	+0.6	[−3.5, +4.5]	0.794	5.8	+0.6	[−3.6, +4.8]
Gemma 4B	**−5.9**	[−10.8, −0.9]	0.018	7.0	**−6.0**	[−11.0, −1.0]
DS-V4.1-Flash	+2.7	[−0.7, +6.0]	0.118	4.7	+2.7	[−0.7, +6.1]
Qwen3.8-Flash	**+7.3**	[+3.7, +11.2]	<0.001	5.4	**+7.3**	[+3.6, +11.2]
GLM-5.3-Flash	+4.7	[−0.5, +9.7]	0.080	7.3	+4.4	[−0.9, +9.5]
Small mean	−2.6	[−6.6, +1.3]	0.188	5.7	−2.7	[−6.8, +1.3]
Larger mean	**+4.9**	[+1.5, +8.3]	0.006	4.9	**+4.8**	[+1.3, +8.3]
$B = 15$						
Qwen 4B	−3.0	[−6.5, +0.6]	0.106	5.1	−3.1	[−6.6, +0.6]
Gemma 4B	**−6.2**	[−10.3, −2.1]	0.003	5.8	**−6.3**	[−10.6, −2.1]
DS-V4.1-Flash	−0.7	[−4.7, +3.2]	0.762	5.5	−0.7	[−4.6, +3.4]
Qwen3.8-Flash	+0.3	[−3.5, +4.3]	0.894	5.5	−0.2	[−3.9, +3.7]
GLM-5.3-Flash	+0.8	[−3.5, +5.0]	0.726	6.2	+0.7	[−3.6, +4.9]
Small mean	**−4.6**	[−7.9, −1.3]	0.009	4.7	**−4.7**	[−8.2, −1.2]
Larger mean	+0.2	[−3.3, +3.7]	0.922	5.0	−0.1	[−3.5, +3.6]
RQ2: fine-tuned Qwen 4B						
SFT w/ advice − statistician ($B = 10$)	+1.4	[−2.2, +5.0]	0.461	5.2	+1.4	[−2.3, +5.1]
SFT w/ advice − SFT w/o advice	**+16.5**	[+13.0, +20.1]	<0.001	5.1	**+17.0**	[+13.4, +20.7]
RQ2: questions asked, LLM with advice − statistician						
$B = 15$						

Comparison	Diff.	95% CI	p	MDE	Diff.	95% CI
Larger mean[q]	**−1.38**	[−1.65, −1.12]	<0.001	0.38	**−1.37**	[−1.65, −1.10]
RQ2: LLM with advice − statistician with as many questions (Figure 3)						
$B = 10$						
Qwen 4B	+0.7	[−3.3, +4.6]	0.748	5.8	+0.7	[−3.4, +4.8]
Gemma 4B	**−5.9**	[−10.8, −0.9]	0.018	7.0	**−6.0**	[−11.0, −1.0]
DS-V4.1-Flash	**+3.3**	[+0.2, +6.7]	0.048	4.7	+3.4	[+0.0, +6.8]
Qwen3.8-Flash	**+8.7**	[+5.0, +12.5]	<0.001	5.5	**+8.7**	[+4.8, +12.6]
GLM-5.3-Flash	**+5.3**	[+0.2, +10.5]	0.047	7.3	+5.1	[−0.2, +10.2]
Small mean	−2.6	[−6.6, +1.4]	0.200	5.7	−2.6	[−6.8, +1.4]
Larger mean	**+5.8**	[+2.4, +9.2]	<0.001	4.8	**+5.7**	[+2.2, +9.1]
$B = 15$						
Qwen 4B	−2.7	[−6.2, +0.8]	0.136	5.0	−2.8	[−6.3, +0.9]
Gemma 4B	**−6.2**	[−10.3, −2.1]	0.003	5.8	**−6.3**	[−10.6, −2.1]
DS-V4.1-Flash	**+4.3**	[+0.5, +8.0]	0.027	5.3	**+4.4**	[+0.7, +8.2]
Qwen3.8-Flash	**+8.2**	[+4.2, +12.3]	<0.001	5.8	**+7.8**	[+3.7, +12.1]
GLM-5.3-Flash	+4.2	[−0.3, +8.7]	0.077	6.4	+4.1	[−0.5, +8.7]
Small mean	**−4.4**	[−7.7, −1.1]	0.010	4.7	**−4.5**	[−8.0, −1.1]
Larger mean	**+5.6**	[+2.2, +8.9]	0.001	4.7	**+5.4**	[+2.1, +8.8]
RQ2: LLM with advice − statistician's own stopping rule at the same mean number of questions						
$B = 10$						
DS-V4.1-Flash	+2.7	[−0.4, +5.8]	0.091	4.4	+2.7	[−0.4, +5.9]
Qwen3.8-Flash	**+7.3**	[+3.8, +10.9]	<0.001	5.1	**+7.3**	[+3.7, +11.0]
GLM-5.3-Flash	+3.4	[−1.4, +8.1]	0.158	6.7	+3.1	[−1.7, +7.9]
Larger mean	**+4.5**	[+1.4, +7.5]	0.003	4.4	**+4.4**	[+1.2, +7.5]
$B = 15$						
DS-V4.1-Flash	**+4.8**	[+0.6, +8.9]	0.021	5.8	**+4.9**	[+0.6, +9.1]
Qwen3.8-Flash	**+8.3**	[+4.4, +12.4]	<0.001	5.7	**+8.0**	[+4.1, +12.1]
GLM-5.3-Flash	+0.8	[−3.5, +5.0]	0.696	6.1	+0.7	[−3.6, +4.9]
Larger mean	**+4.7**	[+1.4, +7.9]	0.006	4.6	**+4.5**	[+1.3, +7.8]
RQ3: wrong diagnosis − correct advice						
$B = 15$						
Qwen 4B	**−7.8**	[−9.8, −6.0]	<0.001	2.7	**−8.0**	[−9.9, −6.2]
Qwen 4B, fine-tuned	**−5.9**	[−7.2, −4.6]	<0.001	1.9	**−6.0**	[−7.4, −4.7]
DS-V4.1-Flash	−0.8	[−3.8, +2.2]	0.635	4.4	−0.9	[−3.9, +2.4]
Qwen3.8-Flash	**−3.8**	[−7.0, −0.8]	0.017	4.4	**−3.7**	[−6.8, −0.7]
GLM-5.3-Flash	−1.7	[−4.5, +1.2]	0.267	4.0	−1.9	[−4.8, +1.0]
Larger mean	**−2.1**	[−4.0, −0.3]	0.028	2.7	**−2.2**	[−4.1, −0.2]
RQ3: wrong questions − correct advice						
$B = 15$						
Qwen 4B	**−2.7**	[−4.5, −0.8]	0.004	2.6	**−2.8**	[−4.7, −0.8]
Qwen 4B, fine-tuned	**−16.4**	[−20.0, −13.0]	<0.001	5.0	**−16.9**	[−20.5, −13.4]
DS-V4.1-Flash	**−4.2**	[−7.8, −0.7]	0.027	5.2	**−4.6**	[−8.3, −0.9]
Qwen3.8-Flash	**−6.3**	[−10.3, −2.5]	0.001	5.5	**−6.0**	[−9.9, −2.2]
GLM-5.3-Flash	−2.5	[−5.3, +0.2]	0.081	4.0	−2.4	[−5.3, +0.3]
Larger mean	**−4.3**	[−6.7, −2.0]	<0.001	3.4	**−4.3**	[−6.7, −2.0]
RQ3: wrong diagnosis − no advice						
$B = 15$						
Qwen 4B	**+30.7**	[+26.6, +34.7]	<0.001	5.8	**+31.4**	[+27.3, +35.5]
DS-V4.1-Flash	**+28.0**	[+22.5, +33.5]	<0.001	7.8	**+29.1**	[+23.6, +34.5]
Qwen3.8-Flash	**+22.5**	[+17.2, +27.8]	<0.001	7.6	**+23.3**	[+17.9, +28.7]
GLM-5.3-Flash	**+24.8**	[+19.8, +30.0]	<0.001	7.2	**+25.5**	[+20.4, +30.6]

Comparison	Diff.	95% CI	p	MDE	Diff.	95% CI
Larger mean	**+25.1**	[+20.8, +29.5]	<0.001	6.2	**+26.0**	[+21.6, +30.3]
RQ3: wrong questions − no advice						
$B = 15$						
Qwen 4B	**+35.8**	[+31.2, +40.3]	<0.001	6.4	**+36.6**	[+32.1, +41.1]
DS-V4.1-Flash	**+24.7**	[+19.7, +29.7]	<0.001	7.1	**+25.3**	[+20.4, +30.4]
Qwen3.8-Flash	**+20.0**	[+14.5, +25.5]	<0.001	7.8	**+21.1**	[+15.6, +26.7]
GLM-5.3-Flash	**+24.0**	[+19.0, +29.2]	<0.001	7.3	**+25.0**	[+19.9, +30.3]
Larger mean	**+22.9**	[+18.8, +27.1]	<0.001	5.8	**+23.8**	[+19.8, +27.9]
RQ3: all wrong − no advice						
$B = 15$						
Qwen 4B	**−20.2**	[−23.2, −17.1]	<0.001	4.4	**−21.4**	[−24.4, −18.5]
DS-V4.1-Flash	**−26.7**	[−31.8, −21.8]	<0.001	7.1	**−27.2**	[−32.1, −22.3]
Qwen3.8-Flash	**−23.0**	[−27.3, −18.7]	<0.001	6.2	**−24.0**	[−28.2, −19.6]
GLM-5.3-Flash	**−16.2**	[−20.5, −12.0]	<0.001	6.0	**−17.0**	[−21.3, −12.9]
Larger mean	**−21.9**	[−25.2, −18.8]	<0.001	4.5	**−22.7**	[−25.8, −19.7]
Section 4: larger models, accuracy when agreeing − when disagreeing with the statistician						
$B = 10$	**+36.3**	[+23.5, +48.4]	<0.001	17.8	**+37.8**	[+24.5, +50.2]
$B = 15$	**+32.7**	[+18.0, +47.3]	<0.001	21.1	**+34.2**	[+18.8, +50.2]
Section 4: share of all errors in disagreeing interviews (%)						
$B = 10^{\%}$	16.2	[11.4, 21.5]	–	–	16.5	[11.4, 22.0]
$B = 15^{\%}$	13.4	[8.7, 18.8]	–	–	13.5	[8.5, 19.2]

Differences in accuracy points unless marked, rounded as described in Section 3; ^{q}: in questions; $^{\%}$: share of errors in percent. **Bold**: $p < 0.05$. MDE: minimum detectable effect (80% power). Means are over the models listed, per patient. Statistician accuracy: 69.0% at $B = 10$ and 78.7% at $B = 15$ (70.4% and 80.3% without the unlabeled conditions).

3 HOW EACH QUANTITY IS COMPUTED

Quantity	What it tells	How it is computed
Difference	How much more (or less) accurate one condition is, in points.	Per patient: accuracy averaged over that patient's interviews (8 for Qwen 4B and Gemma 4B, 2 for each larger model; the statistician is deterministic), minus the same for the other condition. Then the mean over the 300 evaluation patients.
95% confidence interval	How far the difference could move with another sample of patients.	Patient-paired bootstrap: draw 300 patients with replacement, average their differences, repeat 10,000 times (seed 0); take the 2.5th and 97.5th percentiles.
p-value	Whether the difference could be zero.	Twice the share of the 10,000 bootstrap averages on the other side of zero. Significant if $p < 0.05$, i.e. the interval excludes zero. Two methods match when the difference is not significant.
MDE	The smallest true difference the evaluation would detect in 80% of repetitions.	$2.80 \times$ SE, with SE the standard deviation of the bootstrap averages ($2.80 = 1.96 + 0.84$).
Statistician reading the LLM's answers	What the LLM's questions are worth, apart from its diagnosis.	The statistician's most likely condition given exactly the answers the LLM collected. Random questions: the same number drawn from the same menu, averaged over 5 seeds. Own questions: the statistician's information-gain questions, stopped after the same number.
As many questions	The comparison of the paper's Figure 3.	In each interview, the statistician asks exactly as many of its own questions as the LLM asked, ignoring its 0.9 threshold, and diagnoses the most likely condition.
Own stopping rule	The same comparison with the statistician's usual rule.	The statistician's accuracy at budgets 2–15 with its 0.9 threshold, linearly interpolated per patient at the LLM's mean number of questions.
Review signal	Whether disagreement marks interviews worth a second look.	Interviews of the larger models with advice, split by whether the LLM's diagnosis equals the statistician's most likely condition given the same answers; accuracy in agreeing minus disagreeing interviews. The bootstrap resamples patients with all their interviews.
Rounding	How the numbers here relate to those in the paper.	The paper prints accuracies and numbers of questions with one decimal, and every number in its text that can be computed from its tables and figures is computed from these printed values (e.g., 1.3 fewer questions $= 11.7 - 10.4$ in Table 1; gains of 26.3 to 50.7 points from Table 1). Here every difference is computed from the unrounded per-patient values and then rounded, so it can differ by 0.1 from the difference of two printed values (e.g., Gemma 4B, advice − no advice at $B = 15$: +50.6 here, $72.5 - 21.8 = 50.7$ in the paper). The difference in questions is shown with two decimals ($11.74 - 10.36$).
Without unlabeled conditions	Whether the conditions with no labeled patients drive a result.	6 of the 300 evaluation patients (2%) have such a condition (Ebola); each comparison is repeated without them.